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Forensic Science 35

Assignment Booklet 3

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Summary

	Total Possible Marks	Your Marks
Lesson 1	24	
Lesson 2	22	
Lesson 3	21	
Lesson 4	23	
TOTAL	90	

Teacher's Comments

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Forensic Science 35

Assignment Booklet 3

Arson and Explosives



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Forensic Science 35

Assignment Booklet 3

Assessment

For successful completion of this course, you must

- 1. Complete all questions in each Assignment Booklet to the best of your ability.**
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- 2. Achieve at least 40% on the final exam.**
In accordance with Alberta Distance Learning Centre policy, the final exam is worth 50% of the final course mark.
- 3. Achieve a final course mark of at least 50%.**

Process

- **Read the course material** before working on the activities in this Assignment Booklet. The *Forensic Science 35 Student Module Booklets* contain the information necessary for successful completion of your work.
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Module 3, Lessons 1 - 4**Total
24****Lesson 1 Assignment: Combustion and Arson****Value
10****Part A: Multiple-Choice Questions**

Decide which of the choices best completes the statement or answers the question.
Print the letter that corresponds to your choice on the blank next to the question.

- A** 1. What three things must be present to start a fire?
- A. Fuel, oxidant, and a heat source
B. Air, nitrogen, and a hydrocarbon
C. Oxygen, water, and an ignitable liquid
D. Accelerant, energy, and carbon dioxide
- D** 2. Which of the following substances is commonly used by arsonists to start fires?
- A. Oxidant
B. Nitrogen
C. Decelerant
D. Accelerant
- A** 3. Two types of energy produced in a combustion reaction are
- A. heat and light
B. water and steam
C. butane and gasoline
D. oxygen and carbon dioxide
- C** 4. Which of the combinations identifies the oxygen level in the atmosphere and the level of oxygen needed to sustain a fire?

Row	Atmosphere O ₂ Levels	Levels of O ₂ Needed for Fire
A.	0.93%	less than 0.04%
B.	16%	less than 78%
C.	21%	greater than 16%
D.	78%	greater than 21%

- C** 5. The ignition range is the temperature at which a combustion reaction can
- A. produce flames that are long-lasting
 - B. give off enough energy to create steam
 - C. sustain itself without adding more fuel or oxygen
 - D. support the burning of oxygen without the release of carbon dioxide
- A** 6. Which of the following relationships describes a factor that affects the speed of a combustion reaction?
- A. Higher temperatures increase the speed of the reaction.
 - B. Lower temperatures increase the speed of the reaction.
 - C. Higher nitrogen levels increase in the speed of the reaction.
 - D. Lower nitrogen levels slow the speed of the reaction.
- C** 7. Where are the majority of arson fires set?
- A. Homes
 - B. Vehicles
 - C. Outdoors
 - D. Businesses
- B** 8. The majority of arsonists are motivated by
- A. profit
 - B. revenge
 - C. vandalism
 - D. excitement
- B** 9. The usual type of crime that arsonists tend to try to conceal by setting a fire is
- A. murder
 - B. burglary
 - C. trespassing
 - D. sexual assault

- A** 10. The majority of arsons are committed by
- A. males under the age of 18
 - B. females under the age of 35
 - C. males between 18 and 35 years of age
 - D. females between 18 and 35 years of age

Value
3

Part B: Numeric-Response Questions

Use the following information to answer the next question.

Various Definitions Related to Combustion and Arson

1. Energy given off as a product
2. Temperature above which an ignitable liquid will ignite
3. A liquid or solid fuel source that increases the rate of combustion
4. Temperature at which ample amounts of fuel produce flames and high heat

①

1. Match the definitions numbered above with the terms stated below.

Definition:	<u> 3 </u>	<u> 2 </u>	<u> 1 </u>	<u> 4 </u>
Term:	accelerant	flash point	exothermic reaction	flammable range

Use the following information to answer the next question.

Various Descriptions of Arson Motives

1. Usually committed by adults rarely by juveniles
2. Targets are large and usually outdoors.
3. Disagreements or jealousy are common motives for arson
4. Usually committed by two or more male juveniles

①

2. Match the descriptions numbered above with the terms stated below.

Definition:	<u> 3 </u>	<u> 1 </u>	<u> 2 </u>	<u> 4 </u>
Arson Motive:	revenge	profit	excitement	vandalism

Use the following information to answer the next question.

Arson Motives

1. Profit
2. Revenge
3. Vandalism
4. Excitement

- ① 3. The ranking of the arson motives above from most common to least common is 2 , 4 , 3 , and 1 .

Value
11

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

- ③ 1. List three ways that an accelerant used by an arsonist may be detected by a fire investigator.

Any three of the following are acceptable for 3 marks

- | | |
|--|--|
| • <i>Unusual quantities of accelerant (especially when located in an unusual area)</i> | • <i>Rapid rate of fire growth</i> |
| • <i>Unusual types of fuel (especially when located in an unusual area)</i> | • <i>Size of area damaged by fire - large distinctive burn patterns in fire debris</i> |

- ② 2. a. Provide examples of a liquid fuel source and a solid fuel source often used by arsonists.

Any one of the following are acceptable

- *Liquid fuel source - gasoline, diesel, kerosene, turpentine, butane 1 mark*
- *Solid fuel source - wicker, foam, paper, wood 1 mark*

- ② b. Explain why an arsonist would choose each of these fuel sources.

Any one of the following are acceptable

- *Liquid fuel source - readily available at stores, easy to use, fast, spreads easily 1 mark*
- *Solid fuel source - readily available in homes or buildings, may possibly prevent suspicion, increase rate of fire growth 1 mark*

②

3. Describe two differences in the behaviour of an adult arsonist and the behaviour of a juvenile arsonist.

Any two of the following are acceptable for 2 marks

Differences- Adult

- *Adult arsonists tend to work alone*
- *Adult arsonist's motive is usually revenge or profit*
- *Adult arsonist's target is usually buildings*

Differences - Juvenile

- *Juvenile arsonists often work in pairs or groups*
 - *Juvenile arsonist's motive is usually vandalism or excitement*
 - *Juvenile arsonist's target is usually schools, abandoned buildings*
-

②

4. Describe one motive that is characteristic of young arsonists and describe how this behaviour can be stopped.

Any one of the following are acceptable for

- *Motives of young arsonists include boredom, curiosity about fire, accidents, peer pressure, or expressions of anger and stress. 1 mark*
 - *Child arsonist's often will stop their behaviour when counselled by an adult. Recognizing and identifying children who have problems with setting fires deliberately is important because often their behaviour can be stopped through counselling. 1 mark*
-

**Total
22****Lesson 2 Assignment: Investigating Arson Fires****Value
10****Part A: Multiple-Choice Questions**

Decide which of the choices best completes the statement or answers the question.
Print the letter that corresponds to your choice on the blank next to the question.

- D** 1. Which of the following terms best describes the type of evidence usually collected in arson cases?
- A. Concealed
 - B. Combustion
 - C. Conclusive
 - D. Circumstantial
- C** 2. To classify a fire, investigators first try to
- A. interview eyewitnesses
 - B. establish the cause of the fire
 - C. locate the origin of the fire
 - D. question the owner of the property
- A** 3. After investigators determine who had access to the arson scene, they will
- A. ask these individuals for alibis
 - B. search the homes of these individuals
 - C. give each individual a lie detector test
 - D. take the fingerprints of these individuals
- C** 4. Upon whom will fire investigators focus first when trying to establish proof of motive?
- A. 911 caller
 - B. Eyewitnesses
 - C. Owner of property
 - D. Relatives of owner

B

5. In which of the following materials or locations does fire tend to burn longest and thus cause the most damage?

A. Carpet
B. Point of origin
C. Wood floor joists
D. Nonporous floors

A

6. The bottom of a classic V burn pattern represents the location of the ---i--- severe fire damage on an ---ii--- surface. The statement above is best completed by the information in row

Row	<i>i</i>	<i>ii</i>
A.	most	vertical
B.	most	horizontal
C.	least	vertical
D.	least	horizontal

D

7. Which of the following observations is evidence that hydrocarbons are used in an accelerant?

A. Distinct V patterns and dark soot
B. Circular burn patterns and melted material
C. Little physical damage and superficial scorching
D. Extensive physical damage and distinct dark-coloured patterns

B

8. Which of the following accelerants tends to cause the least physical damage (superficial scorching)?

A. Diesel
B. Alcohol
C. Gasoline
D. Kerosene

B

9. A device that consists of a series of glass sampling tubes lined with absorbent material which collects vapours is a

A. gas chromatography unit
B. portable arson sampler
C. photoionization detector
D. solid-phase microextraction unit

- A** 10. Which of the following pairs of technologies and/or techniques is the best way to identify unknown accelerants in an arson fire?
- A. Gas chromatography and mass spectrometry
 - B. Portable arson sampler and metal oxide sensors
 - C. Accelerant detection dogs and criminal profiling
 - D. Photoionization detectors and solid-phase microextraction

Value
2

Part B: Numeric-Response Questions

Use the following information to answer the next question.

Indoor Fire Scene Investigation Tasks

1. Evidence collection area is created.
2. A detailed diagram of the site is drawn.
3. Investigators try to determine the fire's point of origin.
4. Access to the fire scene is controlled by the police and the fire department.

①

1. The correct sequence of tasks that arson experts follow when investigating a fire scene include **4** , **1** , **3** , and **2** .

Use the following information to answer the next question.

Description of Burn Patterns Found at an Arson Scene

1. Produced after a liquid accelerant is poured on a carpet and set on fire
2. Ignitable liquid fuel that has absorbed and burned into wooden floor joists
3. Produced as a fire moves up a vertical surface
4. Dark-coloured burn patterns produced when a liquid accelerant burns on a nonporous surface

①

2. Match each description numbered above with its correct burn pattern stated below.

Definition:	<u> 3 </u>	<u> 1 </u>	<u> 2 </u>	<u> 4 </u>
Burn Pattern:	classic V	doughnut	rundown	ignitable liquid pour

Value
10

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

③

1. Describe three reasons arson fires are difficult to solve.

Any three of the following answers are acceptable for 3 marks

- **Usually done at night**
- **Committed away from eyewitnesses**
- **Evidence often destroyed by fire**
- **Evidence often circumstantial**

③

2. Describe three forms of evidence that indicate that a fire was caused by an arsonist.

Any three of the following answers are acceptable for 3 marks

- **Multiple points of origin**
- **Point of origin near a good supply of oxygen (such as an open window)**
- **Evidence that fire burned quickly and for a relatively long time**
- **Empty fuel cans or other evidence of the use of accelerants**
- **Unusual odours caused by the use of accelerants**

“The most common accelerant used in arson fires by far is gasoline because it is easy to obtain and does not cause any suspicions when being purchased.”

①

3. a. What can fire investigators use to identify gasoline residue at a fire scene?

Photo Ionization Detector 1 mark

①

- b. Identify a laboratory method that can confirm that evidence from a fire scene contains gasoline.

Any one of the following answers are acceptable 1 mark

- **Gas chromatography**
- **Mass spectrometry**
- **Metal oxide sensor**

②

4. Why are gas chromatography and mass spectrometry often both used when analyzing unknown evidence from an arson scene?

Most of the accelerant residue found in fire debris involves complex mixtures. 1 mark

Mass spectrometry can identify only one compound at a time and not a mixture. However, gas chromatography is very effective at separating the individual compounds in an unknown mixture. 1 mark

Total
23

Lesson 3 Assignment: Explosives

Value
10

Part A: Multiple-Choice Questions

Decide which of the choices best completes the statement or answers the question. Print the letter that corresponds to your choice on the blank next to the question.

- B** 1. What causes the casing of an explosive device to burst and fragment?
- A. An increase in temperature outside the casing causes melting.
 - B. A build-up of gases inside the casing leads to an increase in pressure.
 - C. A build up of shock waves inside the casing leads to an increase in friction.
 - D. A decrease in pressure outside the casing causes shock waves to be generated.
- B** 2. Which of the following characteristics describes a point detonating fuse?
- A. Requires light to ignite
 - B. Will combust upon impact
 - C. Explodes after exposure to metal
 - D. Detonates after a set period of time
- D** 3. Usually, most of the time and effort spent creating an explosive device is spent on the
- A. casing
 - B. oxidizer
 - C. fuel source
 - D. ignition source
- B** 4. Which of the following substances is an example of a pure explosive compound?
- A. Dynamite
 - B. Nitroglycerin
 - C. Flash powder
 - D. Black powder

- A** 5. To start a fire, substance must be exposed to _____. To detonate an explosion, substances must be exposed to _____.

A. heat; heat or shock
B. heat or shock; heat
C. heat; sound
D. sound; heat
- D** 6. Which of the following devices is an example of a high explosive?

A. Flare
B. Car bomb
C. Pipe bomb
D. Military warhead
- D** 7. Which of the following pairs of scientific techniques does an EGIS system utilize?

A. X-ray analysis and television monitors
B. Oxidation and combustion
C. Heat and mechanical shock
D. Gas chromatography and mass spectrometry
- B** 8. Which of the following technologies or techniques cannot be used to identify the type of explosive compounds within a bomb?

A. X-ray machines
B. Bomb detection dogs
C. Ion mobility spectrometry
D. Erieye ground interface segment
- A** 9. Which of the following technologies or techniques is most accurate when used to identify an explosive compound?

A. EGIS system
B. X-ray machines
C. Ion mobility spectrometry
D. Remote mechanical investigator

- D** 10. A method that may be used to safely remove and/or detonate an explosive device is
- A. the EGIS system
 - B. a bomb detection dog
 - C. the ion mobility spectrometry
 - D. the remote mechanical investigator

Value
2

Part B: Numeric-Response Questions

Use the following list to answer the next question.

Description of Explosives and Parts of Explosives

1. Ignites an explosive using wires or radio waves
2. A length of cord filled with combustible material
3. Relatively insensitive to heat, friction, or temperature
4. Generates shock waves that have speeds of 2300 metres per second

①

1. Match each description numbered above with its correct term stated below.

Definition:	<u> 2 </u>	<u> 4 </u>	<u> 3 </u>	<u> 1 </u>
Term:	fuse	low explosive	high explosive	remote detonator

Use the following information to answer the next question.

Descriptions of Explosions and Fires

1. Requires oxygen
2. A detonation
3. Produces heat
4. Produces nitrogen
5. Fuel contains a relatively large amount of energy that is released slowly
6. Fuel contains a relatively small amount of energy that is released quickly

①

2. Four descriptions of an explosion include **1** , **2** , **3** , and **6** .

(Record your answer from lowest to highest numeric value.)

Value
9

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

②

1. Describe a fuse, and explain why some explosive devices require one.

A fuse is a length of cord either filled with combustible material or made from combustible material. (1 mark)

Some devices require a fuse to produce heat to cause a reaction between the compounds in the metal casing resulting in an explosion. (1 mark)

②

2. Identify and describe a major disadvantages of using both the X-ray machine and the metal detector to detect explosives.

Both methods require visual confirmation by a person if a positive reading is made. (1 mark)

Operators of both these devices rely heavily on visual analysis of objects. Thus, explosives that are cleverly hidden may be missed. (1 mark)

②

3. Could the EGIS system replace the use of X-ray machines or metal detectors in airports? Explain.

No. The EGIS system takes a long period to confirm its findings. This would disrupt airport scheduling.

③

4. Identify three features of the device used to destroy Pan Am flight 103.

Any three of the following answers are acceptable for 3 marks

- ***plastic explosive***
- ***made from Semtex***
- ***hidden in a suitcase***
- ***time-delay ignition/detonation***
- ***about 450 grams in size***
- ***unaccompanied piece of luggage***
- ***hidden inside a radio cassette recorder***

Total
21

Lesson 4 Assignment: Crime Case Studies Involving Arson and Explosives

Value
10

Part A: Multiple-Choice Questions

Decide which of the choices best completes the statement or answers the question. Print the letter that corresponds to your choice on the blank next to the question.

Historical Crime Case Study #1: A Deadly Serial Arsonist

- A** 1. Five firefighters died in the *Esperanza Fire* trying to save a
- A. vacant, partially built home
 - B. house containing a family of four
 - C. animal facility containing numerous dogs
 - D. gas station containing explosive hydrocarbons
- C** 2. What occurred after Raymond Lee Oyler was arrested?
- A. Oyler confessed to setting the *Esperanza Fire*.
 - B. Oyler's fingerprints were found at the Old Fire site.
 - C. No new arson fires occurred in the area near Oyler's home.
 - D. Several new arson fires occurred in the area near Oyler's home.
- D** 3. Serial arsonists are often predictable in that they tend to
- A. create the same type of diversion
 - B. use the same type of getaway vehicle
 - C. start their fires at the same time of day
 - D. use the same type of incendiary device
- C** 4. Which of the following types of incendiary devices caused the *Old Fire* and the *Esperanza Fire*?
- A. Jar filled with liquid fertilizer
 - B. Wine bottle filled with kerosene
 - C. Matches wrapped around a cigarette
 - D. Box containing two sticks of dynamite

- A** 5. Which of the following types of evidence linked Raymond Lee Oyler directly to two of his arson fires?
- A. DNA
 - B. Footprints
 - C. Fingerprints
 - D. Inorganic fibres

Historical Crime Case Study #2: The Oklahoma City Bombing

- D** 6. Many children died in the *Oklahoma City Bombing* because the bomb was placed
- A. near a playground
 - B. beside a school bus
 - C. close to a play school
 - D. under a daycare centre
- C** 7. The explosive device used in the *Oklahoma City Bombing* was hidden in a
- A. stolen car
 - B. parcel box
 - C. rental truck
 - D. plastic bucket
- D** 8. Which compound in the bomb created by Timothy McVeigh and Terry Nichols serve as the oxidizing agent?
- A. Tovex
 - B. TNT
 - C. Nitromethane
 - D. Ammonium nitrate
- C** 9. Why was Timothy McVeigh arrested so soon after the bombing?
- A. He stole a car.
 - B. He turned himself in.
 - C. His car had no licence plate.
 - D. The rental car agency phoned police.

- D 10. If the detonation device devised by Timothy McVeigh failed, he intended to
- A. commit suicide
 - B. turn himself in to police
 - C. trigger a second time-delay detonator
 - D. ignite extra explosives with a handgun

Value
2

Part B: Numeric-Response Questions

Historical Crime Case Study #1: A Deadly Serial Arsonist

Use the following information to answer the next question.

Various Pieces of Evidence

1. DNA
2. Fingerprints
3. Testimony from Oyler's boss
4. Footprints left at two of the arson scenes
5. Testimony from Oyler's cousin and girlfriend
6. Surveillance video from casino and gas station
7. Eyewitness statement from gas station attendant
8. Videotape of Oyler leaving the scene of one of the fires

- ① 1. Four pieces of evidence that may help convict Raymond Lee Oyler of arson include 1 , 5 , 6 , and 8 .

(Record your answer in ascending order.)

Descriptions of Explosions and Fires

Use the following information to answer the next question.

Events that occurred in the Oklahoma City Bombing

1. The explosive device is built.
2. Timothy McVeigh is arrested.
3. Terry Nichols turns himself in.
4. The gateway vehicle is placed near the target building.
5. The explosive device is placed near the target building.

- ① 2. The correct sequence of events in the Oklahoma City Bombing is 4 , 1 , 5 , 2 , and 3 . *The assignment booklet has only 4 blanks and should have 5. This will be corrected in the next printing*

Value
11

Part C: Written-Response Questions

Provide appropriate answers to the following. Explanations are best given in sentences.

Historical Crime Case Study #1: A Deadly Serial Arsonist

3

1. Describe how Raymond Lee Oyler was able to set the wildfires without being seriously burned.

- **Oyler used a device that was time-delayed. (1 mark)**
- **It consisted of five to seven paper or wood matches attached around a Marlboro cigarette with duct tape or a rubber band. When the cigarette was lit, it burned slowly until it reached the matches, ignited them, and started a brush fire in about 10 minutes. (2 marks)**

2

2. What have some suggested could have prevented the death of the five firefighters in the *Esperanza Fire*?

- **Firefighters should not be sent to protect vacant houses or isolated individual homes. (2 marks)**

Historical Crime Case Study #2: The Oklahoma City Bombing

3

3. Was the bomb used to destroy the *Alfred P. Murrah Federal Building* a high explosive or a low explosive, or did it have traits of both? Explain your choice.

- **The bomb exhibited traits of both low and high explosives. 1 mark**
- **It had traits of a low explosive because it used some easy-to-find materials, especially ammonium nitrate (fertilizer) and nitromethane (fuel). (1 mark)**
- **It had traits of a high explosive because it was a large bomb, produced powerful shock waves, and used compounds used in mining, especially Tovex and ANFO. (1 mark).**

③

4. Michael Fortier knew about the Oklahoma City Bombing, yet he failed to inform authorities. Explain why he received a light sentence despite this.

Michael Fortier agreed to testify against McVeigh and Nichols in exchange for a modest sentence and immunity for his wife. (3 marks)

End of Assignment Booklet 3

